

Fachhochschule Hagenberg



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Lipase Activity in Milk

Hydrolysis of Milk Fat by Pancreatic Lipase

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1 Background

1.1 Milk Fat and Digestion

Milk is an oil-in-water emulsion: fat is present as microscopic globules coated by a protein-phospholipid membrane (the milk fat globule membrane, MFGM). Typical cow's milk contains about 3-4% lipids (mostly triacylglycerols), with hundreds of different fatty acids; roughly 11% of these are short-chain acids ($\approx C4-C10$), almost half of which is butyric acid. In vivo, many short and medium-chain fatty acids from milk are absorbed directly by intestinal cells and shuttled via the portal blood to the liver for metabolism, often without needing full luminal hydrolysis. However, pancreatic lipase (the digestive enzyme from the pancreas) normally cleaves dietary triglycerides into glycerol and free fatty acids. In our test-tube model, the very fine dispersion of milk fat globules (large surface area) and the presence of many short-chain glycerides make milk an excellent substrate for rapid enzymatic hydrolysis. When lipase acts on milk fat, it releases free fatty acids (and glycerol), acidifying the mixture.

1.2 Principle of the Indicator

The experiment uses phenolphthalein indicator in an alkaline milk solution. Phenolphthalein is pink in sufficiently alkaline solutions (around pH 10-12) and turns colorless when the pH falls below about 8.3. By adding a measured amount of NaOH to the milk (turning it pink), we create an initial alkaline condition. As active lipase cleaves triglycerides to free fatty acids, the pH of the mixture drops. This produces a visible color change from pink toward colorless once pH drops below the indicator threshold. In effect, the duration until loss of pink color (and the pH-versus-time curve) reports the rate of lipase action: rapid acid release will quickly neutralize the base, whereas no acid release (or slow release) will keep the solution pink longer.

1.3 Experimental Rationale

This protocol demonstrates the fundamental principle that pancreatic lipase cleaves milk triglycerides to free fatty acids and glycerol, and that free fatty acids cause a measurable drop in pH (observed via phenolphthalein) when hydrolysis occurs. The contrast with boiled (inactive) enzyme and with low-fat milk reinforces the roles of enzyme activity and substrate availability. The experiment models intestinal fat digestion in vitro, showing that only active lipase can catalyze the reaction, and that more substrate (fat) yields more acid for neutralization.

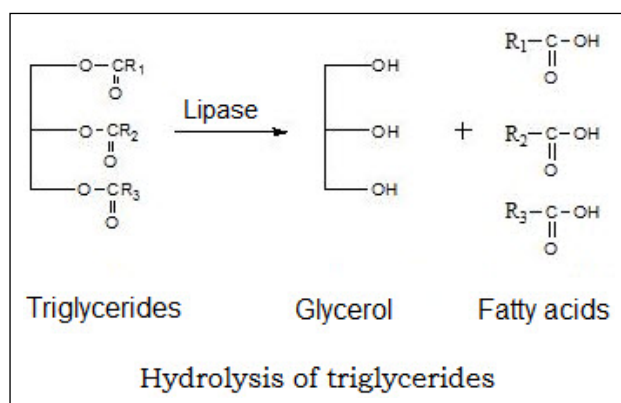


Figure 1: Lipase acts like scissors, cutting a fat molecule (triglyceride) into three fatty acids and one glycerol.

2 Experimental Setup

2.1 Materials

- Pipettes
- Test tubes
- Water bath (37 °C)
- pH meter
- Milk (3.5% fat, and optionally low-fat milk)
- 1M NaOH
- Pancreatic lipase
- 1% phenolphthalein solution
- 0.9% NaCl solution

2.2 Safety Notes

- **NaOH is a strong base. Wear goggles and gloves.**
- **When boiling enzyme solution, point the tube away from yourself and others. Boiling may cause splattering.**

2.3 Preparation

1. Dissolve 50 mg of pancreatic lipase in 10 mL of 0.9% NaCl (shake for 10 min, then let settle).
2. Boil 2 mL of this solution in a test tube using a clamp and Bunsen burner. Allow to cool.

2.4 Setup

Label two test tubes (Tube 1 and Tube 2). For each:

- 5 mL whole milk ($\approx$ 3.5% fat)
- $\sim$ 2 drops 1% phenolphthalein indicator solution
- $\sim$ 50 μ L 1M NaOH (until pink)

Mix gently. The NaOH will alkalize the mixture, turning it pink. Measure and record the initial pH (should be strongly alkaline, $\sim$ pH 10-12).

Add:

- Tube 1: 200-500 μ L boiled (inactivated) lipase solution
- Tube 2: same amount of unboiled (active) lipase solution

Record the exact volumes added and measure pH again.

Incubate both tubes at 37 °C in a water bath. Start timing. Continuously or periodically (e.g. at 30s, 1min, 5min, etc.) observe the color and measure the pH of each tube. The solution in Tube 2 (active enzyme) is expected to lose the pink color over time, while Tube 1 (denatured enzyme) should remain pink. Continue until the color change ceases (typically within 10-60min).

Repeat as needed: For comparison, repeat the experiment using low-fat milk (e.g. skim or 1% fat) and plot all pH-versus-time curves.

3 Data Analysis & Results

3.1 Descriptive Statistics

Table 1 shows a clear difference between the two conditions. The mean pH drop for samples with active (unboiled) lipase was 1.7, more than triple the mean drop of 0.53 for the inactive (boiled) samples. This provides strong initial evidence of enzymatic activity.

	Boiled (Inactive)	Unboiled (Active)
count	65	66
mean	0.53	1.7
std	0.86	1.73
min	-0.87	-0.42
25%	0.11	0.63
50%	0.4	1.13
75%	0.66	2.33
max	4.65	10.01

Table 1: Overall descriptive statistics for the total pH drop. Values are rounded to two decimal places.

Table 2 breaks down the pH drop by the amount of enzyme solution added. While the ANOVA test did not find a statistically significant effect, this table reveals a weak trend: larger volumes of unboiled lipase tend to correspond to a greater average pH drop. For instance, the mean drop for 200 μL was 1.42, while for 600 μL it was 3.3. The high standard deviation in each group likely contributed to the non-significant ANOVA result.

Menge (μL)	Group	N	Mean Drop	Std. Dev.	Max Drop
200	Boiled	11	0.62	1.04	3.67
	Unboiled	12	1.42	2.74	10.01
250	Boiled	1	0.18	-	0.18
	Unboiled	1	0.87	-	0.87
300	Boiled	1	0.7	-	0.7
	Unboiled	1	3.6	-	3.6
350	Boiled	18	0.46	0.6	1.86
	Unboiled	18	0.91	0.85	2.66
400	Boiled	3	0.44	0.12	0.55
	Unboiled	3	2.11	1.38	3.44
450	Boiled	3	0.46	0.36	0.87
	Unboiled	3	2.55	0.99	3.66
500	Boiled	25	0.57	1.11	4.65
	Unboiled	25	2.01	1.55	4.7
600	Boiled	3	0.49	0.29	0.8
	Unboiled	3	3.3	1.98	4.63

Table 2: Descriptive statistics for pH drop grouped by enzyme volume. Values are rounded.

3.2 Visualizations and Statistical Interpretation

The plot of the average pH over time for all groups (Figure 2) clearly visualizes the conclusion from our t-test. The curve for the unboiled enzyme shows a steady decline, indicating continuous acid production, while the boiled enzyme's curve remains nearly flat, confirming its inactivity.

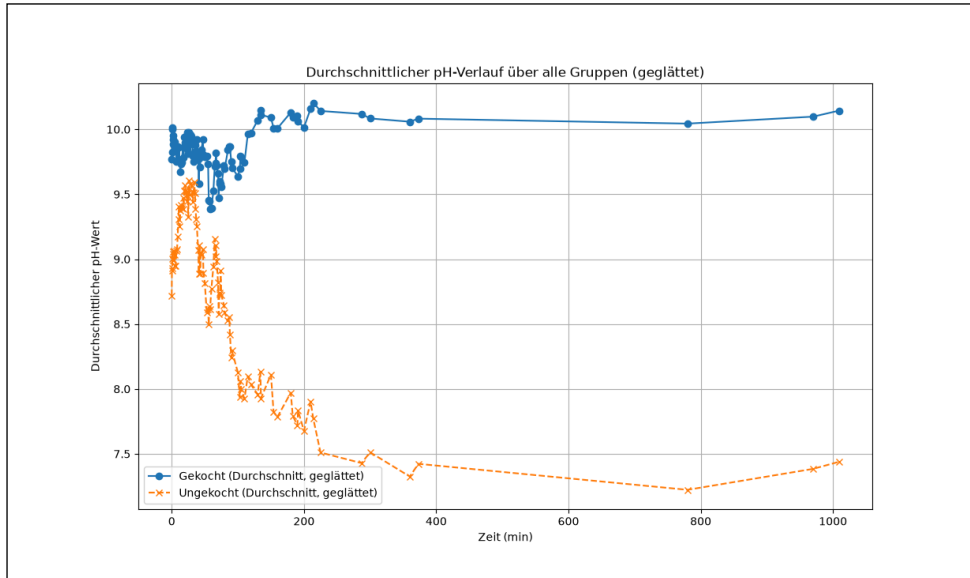


Figure 2: Smoothed average pH vs. time for all groups, comparing boiled (inactive) and unboiled (active) lipase.

The plots in Figure 3 illustrate the dose-response relationship, or lack thereof. For the unboiled enzyme, there is a visible trend where larger amounts of lipase lead to a faster and deeper pH drop. The plot for the boiled enzyme shows no such trend, with all lines clustered together. This visual evidence supports the ANOVA result: while a trend exists for the active enzyme, the high variance within each group prevents the differences from being statistically significant.

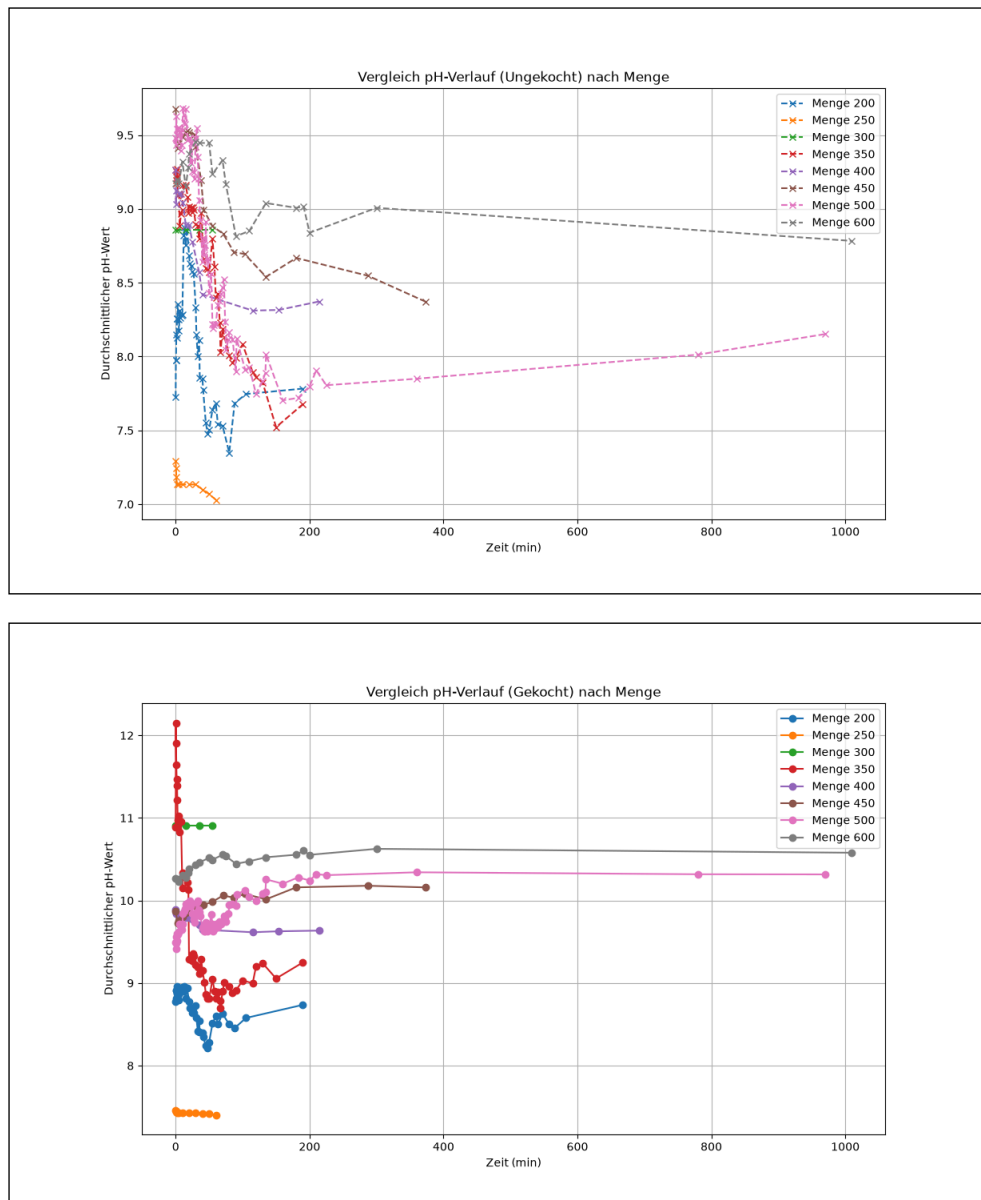


Figure 3: Comparison of smoothed pH drop over time, grouped by enzyme amount.

Top: Unboiled (active) lipase.

Bottom: Boiled (inactive) lipase.

An individual group's plot, like the one in Figure 4, shows the raw data behind the averages, capturing the real-time progress of the experiment.

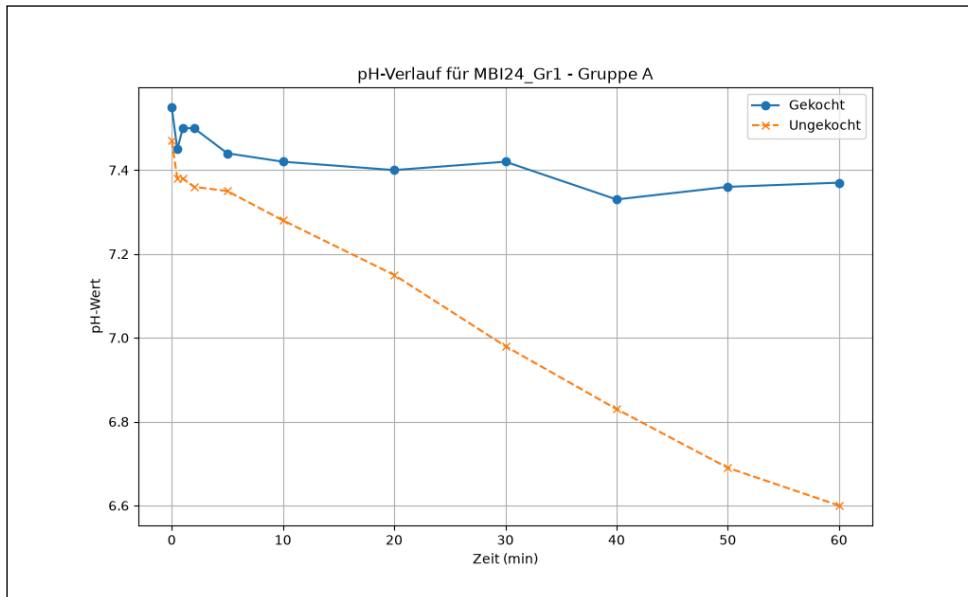


Figure 4: Example of a pH vs. time plot for a single experimental group (MBI24, Group 1A).

3.3 Statistical Tests

To validate these observations, two main statistical tests were performed:

- 1. Paired T-Test (Active vs. Inactive Lipase):** A paired t-test was used to compare the pH drop between the unboiled and boiled samples from the same experimental run. The test yielded a *p-value* of $7.560e-06$, which is far below the standard significance level of 0.05. This result is **highly statistically significant** and confirms that the active, unboiled lipase produces a much greater pH drop than the denatured, boiled lipase.
- 2. ANOVA (Effect of Enzyme Amount):** An ANOVA test was conducted to determine if the amount of lipase added (from 200 μ L to 600 μ L) had a statistically significant effect on the magnitude of the pH drop. The *p-value* from this test was 0.2. Since this value is greater than 0.05, we conclude that there is **no statistically significant difference** in the pH drop between the different amounts of enzyme used in this set of experiments. While a trend is visible in the data and plots, the high variability in the data prevented this trend from being statistically significant.

4 Conclusion

This experiment successfully demonstrated that pancreatic lipase actively hydrolyzes triglycerides in milk, leading to a release of free fatty acids and a measurable drop in pH. The difference in activity between unboiled (active) and boiled (denatured) lipase was statistically highly significant. This confirms that the observed effect is due to the enzyme's catalytic function.

While there was a visible trend suggesting that a higher concentration of lipase leads to a faster reaction, this effect was not statistically significant across the tested range (200-500 μL), likely due to high variability between individual experiments. Overall, the experiment provides a clear and effective model for enzymatic fat digestion.

Appendix

Sources

1. "Roles of Milk Fat Globule Membrane on Fat Digestion and Infant Nutrition - PMC" at <https://pmc.ncbi.nlm.nih.gov/articles/PMC9108948/> (2025-07-10)
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